Climate, sea-level rise, and coastal adaptation in an independent numerical benchmark

Independent scientific-computing investigation

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Abstract

Coastal adaptation costs depend on local exposure as well as global warming. We independently implement a coupled climate, carbon, and sea-level model and connect it to an idealized coastal decision problem. Published equations support reconstruction of substantial physical components, including ocean heat diffusion, four terrestrial carbon reservoirs, a conservative ocean carbon system, glacier and Greenland responses, and Antarctic mass balance. However, the joint calibration, regional sea-level inputs, and coastal data needed to reproduce Figure 1 of Wong et al. (2025) were not recovered. The resulting figure is explicitly a **NON-REPRODUCTION**: it describes three synthetic coastal segments totalling 100 km, not global economic losses. A 4,096-proposal ensemble gives conditional distributions in the published warming categories. Conservation and refinement tests establish numerical consistency; they do not establish agreement with the unavailable calibration or validate synthetic exposure as a representation of the world.

1 Target and limits of inference

Wong et al. [1] condition combined adaptation expenditures and residual coastal losses on 2100 warming relative to 1850–1900. The four categories are $T \leq 1.5$, $1.5 < T \leq 2.5$, $2.5 < T \leq 3.5$, and $T > 3.5$ °C. Their figure reports medians and fifth–95th percentiles in trillion 2010 US dollars, with a 4% annual discount rate and an economic horizon ending in 2150. The main experiment uses SSP4, 12,148 DIVA coastal segments, storm-tide hazards, regional sea-level fingerprints, and nonclimatic local trends. Its 5,000 draws were subsampled from 100,000; bin counts were 202, 2264, 2145, and 389.

Only written scientific sources and independent observations were consulted. No original implementation, processed simulation outputs, posterior samples, prepared model inputs, or figure files were used. Published numerical comparisons below were transcribed from prose rather than digitized. The supplied main paper was read; the supplement could not be retrieved from the journal or its media servers. Therefore the specification is incomplete. The exact discount origin, within-decade accounting, and treatment of initial adaptation cannot be inferred from the phrase "discounted to the year 2150." Our accounting convention is stated separately below.

The forward model and its calibration are distinct scientific objects. Recovering an equation does not recover a joint posterior. The present ensemble is a sensitivity experiment constrained by two historical summaries. It is not the probability distribution used by Wong et al. A synthetic coast permits testing the complete numerical chain while leaving global economic reproduction explicitly unresolved.

2 Physical model

2.1 Land, ocean surface, and ocean heat

We use the DOECLIM equations documented by Kriegler [3], which form the climate component of SNEASY [2]. The ocean is a vertical column of depth $H = 4000$ m, with temperature $\theta(z,t)$. Its surface temperature T_S differs from marine air temperature by factor $b = 1.3$. Land air temperature is T_L and land fraction is $f_L = 0.29$:

$$C_L \dot{T}_L = Q - \lambda_L T_L - \frac{k}{f_L}(T_L - bT_S), \tag{1}$$

$$C_S \dot{T}_S = Q - \lambda_S T_S + \frac{k}{1-f_L}(T_L - bT_S) - F_O, \tag{2}$$

$$\partial_t \theta = \kappa \partial_{zz}\theta, \qquad F_O = -f_O c_V \kappa \partial_z \theta(0,t). \tag{3}$$

The boundary conditions are $\theta(0,t) = T_S(t)$ and $\partial_z\theta(H,t) = 0$; all temperature anomalies initially vanish in 1850. Here $C_L = 0.52$, $C_S = 7.8$ W yr m^{-2} K^{-1}, $c_V = 0.13$ W yr m^{-3} K^{-1}, and $f_O = 0.95$. Time uses 365.25-day years. Diffusivity is converted from cm^2/s to m^2/yr. Global air temperature is $f_L T_L + (1-f_L)bT_S$.

The feedbacks are chosen to give the sampled equilibrium climate sensitivity and the documented equilibrium ratio $T_L/T_S = 1.43$, using $k = 1.59 - 0.31\lambda_L$. This parameterization is checked by inserting the analytic doubled-CO_2 equilibrium into the numerical model. The weighted land–sea heat transport cancels exactly in the global energy budget. Finite volumes discretize the ocean into 40 layers. Crank–Nicolson integration reduces each step to a tridiagonal solve. Total stored heat and radiative loss are independently accumulated to verify conservation.

CO_2 forcing is $Q_C = 3.7\log(C/278)/\log 2$ W/m^2. Other forcing grows linearly from zero in 1850 to 0.56 W/m^2 in 2019 and then stays fixed. This value is motivated by assessed 2019 anthropogenic forcing minus CO_2 forcing [13]; the interpolated history is an assumption. It omits volcanic events and resolved aerosol histories and does not implement nearest-SSP matching. Reference experiments with zero and twice this forcing quantify one consequence of that missing interface.

2.2 Carbon budgets and independent ocean reconstruction

Following the architecture of Ricciuto et al. [4], land has leaf, wood, detritus, and humus carbon inventories L, W, D, S. Net primary production is $N = 60[1 + \beta\log(C/278)]$ Gt C/yr. Temperature multiplies detritus and soil turnover by $r = Q_{10}^{T/10}$. With fluxes $l = 35L/100$, $w = 25W/500$, $d = 55Dr/120$, and $s = 15Sr/1500$, the budgets are

$$\dot{L} = 35N/60 - l, \qquad \dot{W} = 25N/60 - w, \tag{4}$$

$$\dot{D} = l + 0.8w - d, \qquad \dot{S} = 0.2w + 10d/55 - s. \tag{5}$$

The atmosphere gains carbon released by these four pools. Initial reservoir inventories are 100, 500, 120, and 1500 Gt C. These stocks and baseline exchanges use aggregated DCESS II values [7]. They replace unrecovered SNEASY stock constants; they are not evidence of identical configurations. Initial fluxes balance.

The surface ocean equilibrates instantaneously with the atmosphere. Yi's carbonate polynomial [6] gives $C = p(D_0, 18°\text{C})$, where D_0 is surface dissolved inorganic carbon in mol/m^3. Salinity is 35

and alkalinity is 2.35 meq/kg. The fixed-temperature chemistry follows the Ricciuto variant and excludes ocean solubility feedback. The combined surface and atmospheric perturbation inventory is

$$M_0 = 2.12(C - 278) + 12 \times 10^{-15} A_O h_0 (D_0 - D_*), \tag{6}$$

where $A_O = 3.62 \times 10^{14}$ m^2 and $p(D_*, 18) = 278$ ppm. Three deep layers exchange carbon with adjacent layers via $F_i = 12 \times 10^{-15} A_O v_i (D_i - D_{i+1})$ Gt C/yr. Every internal flux enters one inventory and leaves another, so total carbon changes only through the external emissions boundary.

Ocean geometry is independently derived from the Hooss response printed by Tanaka [5]: amplitudes $(0.095, 0.209, 0.253, 0.311, 0.132)$ and decay times $(1.271, 12.17, 59.52, 236.5, \infty)$ years. We replace the fastest mode by instantaneous surface equilibration and realize the remaining modes as a positive four-reservoir diffusion system. Lanczos tridiagonalization of the diagonal decay-rate matrix, starting from square roots of normalized amplitudes, yields a symmetric matrix J. Its positive null vector fixes relative reservoir capacities. Adjacent conductances are $J_{i,i+1}\sqrt{K_i K_{i+1}}$, where K_i is inventory per concentration. This gives depths approximately $(64, 420, 1141, 2390)$ m and exchange velocities $(16.95, 9.08, 6.34)$ m/yr. The resulting depth of 4015 m and thermocline exchange are plausible, but agreement with the original calibrated geometry is unverified.

The carbon ODEs use midpoint Runge–Kutta. A predictor–corrector couples their midpoint concentration to the heat solver. The historical CO_2 boundary is applied at both midpoint and endpoint; omitting its midpoint introduced a first-order lag found and corrected by timestep refinement. Prescribed historical concentrations diagnose compatible emissions rather than validate predicted CO_2. Before 1959 a quadratic curve joins 278 ppm in 1850 to the first observation. NOAA Mauna Loa annual means [12] prescribe 1959–2022. Future emissions begin from an assumed 40.7 Gt CO_2/yr, using Wong's printed annual growth and logistic-decline recurrence. Its discrete decline retains a small positive residual rather than tending exactly to zero; we preserve the printed recurrence and do not silently clamp it.

2.3 Sea-level components

We sum glacier, Greenland, Antarctic, thermal, and land-water contributions. All outputs are referenced to their 1995–2014 mean. The glacier response [8] is

$$\dot{S}_G = \beta_G (T - T_{eq})(1 - S_G/V_G)^{0.78}, \quad T_{eq} = -0.15°\text{C}, \quad V_G = 0.4 \text{ m}. \tag{7}$$

Its temperature reference is 1850–1870. We initialize $S_G = 0$ in 1850, not the calibrated 1990 initialization of the cited model. Greenland obeys $\dot{V} = (V_{eq} - V)/\tau$, with $V_{eq} = 7.8 - 3T$ metres sea-level equivalent and $\tau^{-1} = 0.00073T + 0.00014$ yr^{-1}; T is relative to 1961–1990. We bound V_{eq} to $[0, 8.832]$ and the rate below by zero. These explicit domain choices prevent negative relaxation times. The initial 7.4-m stock is assigned in 1850. Neither ice model is recalibrated here.

Antarctica follows Shaffer's axisymmetric mass balance [9]:

$$\dot{V} = B_{tot} + F, \qquad F = -2\pi R (\rho_w/\rho_i) H_w u, \tag{8}$$

where R is radius, H_w positive grounding-line water depth, and u discharge speed. Surface elevation is $775 - 0.0006R + \sqrt{8.7(R - r)}$ m. Accumulation is $P = 0.35 \exp(0.04 T_A)$ m ice/yr; below runoff height $1471 + 95T_A$ m, ablation is proportional to height deficit with coefficient $0.012\sqrt{P}$. The speed is proportional to $[(1-\alpha) + \alpha((T_O + 1.8)/2.52)^2] H_w^\gamma$, normalized to the published reference geometry. We integrate the analytically integrated surface balance and volume–radius derivative, including

the sea-level derivative of isostatic displacement. The published reference case has $T_A = -18°\mathrm{C}$, $T_O = 0.72°\mathrm{C}$, and sea level zero.

The extracted grounding-depth sign in the written equations is inconsistent with positive water depth for the reference bed. We use sea surface minus bed and recover the reported equilibrium radius and volume. This is an explicit physical interpretation, not a verified transcription of author code. To couple the model, we assume $T_A = -18 + 1.8\Delta T$ and $T_O = 0.72 + \Delta T_{250\mathrm{m}}$, relative to 1961–1990. We omit rapid Antarctic disintegration, regional ice-shelf dynamics, and a cavity-volume correction to the density-based sea-level conversion. These omissions matter especially for the warmest category.

Thermal sea level is $\alpha_V H_O/(4.1 \times 10^6 A_O)$, with heat H_O in joules and constant effective expansion coefficient α_V. Land-water storage contributes an assumed 0.3 mm/yr after 1961. The physical benchmark has no geographic fingerprints or vertical land motion, so its sea level cannot be interpreted as a prediction at a named coastal site.

3 Coastal benchmark and probability measure

The synthetic coastline comprises 40, 35, and 25 km segments with slopes 0.002, 0.005, and 0.010. Assets are 150, 600, and 1800 USD2010/m^2; population densities are 0.0002, 0.002, and 0.008 persons/m^2. These values are hypothetical. Exposed land extends from zero to 10 m elevation. An exponential annual maximum surge has respective scales 0.35, 0.55, and 0.75 m, giving return levels $\sigma \log R$. This is not the GTSR extreme-value hazard.

The flooding damage fraction at elevation z is $\min\{\max[(S + X - z)/2, 0], 1\}$, where S is mean sea-level change and X surge. A seawall eliminates damage until overtopping. Expected loss is integrated analytically over surge and elevation. The five accounts are protection, relocation, permanent capital loss, transient flood/life loss, and wetland services. The full synthetic parameter inventory and equations are recorded in `reports/assumptions.md`. They implement a transparent decision problem inspired by coastal-model descriptions [10, 11]; they are not presented as the published CIAM equations or coefficients.

In 2010 each segment chooses inaction, retreat, or protection and a return level to minimize its first 50-year NPV. Nine active choices cover five retreat and four protection levels. Strategy and return level persist; heights are upgraded in 2060 and 2110. Costs include initial adaptation and are paid at year end for every year 2010–2150, discounted to start-2010 by $(1.04)^{-(y-2010+1)}$. Exposure values grow by 1%/yr in real terms, population is fixed, and present-day flood risk remains in the no-rise counterfactual. These choices differ from the published SSP4, ten-year accounting, and initial adaptation spin-up. A dimensional rescaling from this coast to the world would not recover missing geographic or socioeconomic information.

The independent proposals sample ECS uniformly from 1.5–6°C, diffusivity from 0.5–4 cm^2/s, fertilization from 0.287–0.8, and Q_{10} from 1.2–2.5. A log-uniform 0.5–2 multiplier scales ocean carbon exchange. Emission peaks are discrete-uniform 2023–2100; growth and decline parameters span 0–0.03 and 0.02–0.15, respectively, with log-uniform decline. Glacier response, Antarctic discharge, and thermal expansion also vary over declared ranges. These distributions are assumptions, not a reconstructed posterior. We apply weights

$$w_i = \exp\left\{-\frac{1}{2}\left(\frac{\bar{T}_{2010:2019,i} - 1.06}{0.15}\right)^2 - \frac{1}{2}\left(\frac{\Delta H_{1971:2018,i} - 0.396}{0.10}\right)^2\right\}. \tag{9}$$

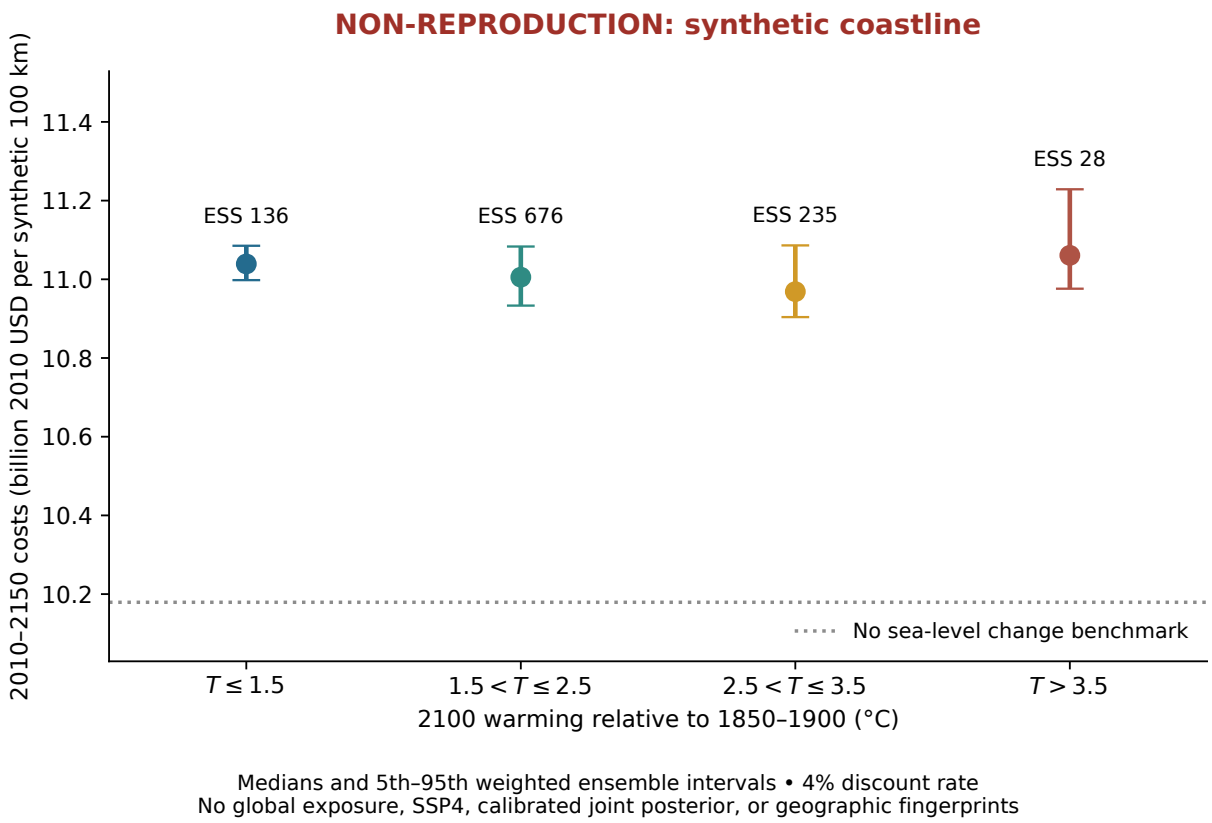


Figure 1: Original, independently simulated **NON-REPRODUCTION** of the conditional plotting task. Points are medians and whiskers fifth–95th percentiles of discounted costs for a synthetic 100-km coastline. Colors identify warming bins; labels report conditional effective sample sizes. The dotted line retains present-day surge risk with no further sea-level change. Units are billion 2010 dollars per synthetic coastline, not the published trillion-dollar global quantity.

The denominators are assumed discrepancy scales, wider than observational errors. The two summaries are treated as independent. Validity filtering excludes any member outside Yi's DIC range before normalizing weights. Conditional intervals invert the weighted empirical CDF. Effective sample size is $(\sum w)^2 / \sum w^2$. Exponential multiplier bootstraps estimate finite-proposal Monte Carlo error, not missing structural uncertainty.

4 Numerical results and comparison

The effective sample size is 1068.6. Maximum retained surface DIC is 2.2902 mol/m^3. The no-rise counterfactual costs 10.179 billion USD2010. Initial adaptation to existing surge hazards therefore explains much of the total cost even in cooler scenarios. The numerical CSVs report all category quantiles, bin weights, effective tail counts, and 400-bootstrap Monte Carlo standard errors. Prefix ensembles of 256, 512, 1024, 2048, and 4096 proposals provide a separate sampling diagnostic. The warmest bin's low effective tail count precludes a claim of precise tail recovery. Enlarging this ensemble would not resolve the absent global inputs. One of 4096 proposals is excluded by the carbonate-domain criterion; its pre-exclusion normalized weight is 8.84×10^{-10}. The warmest

Bin	Draws	ESS	Probability	Median	5th–95th
1	200	135.8	0.121	11.039	10.998–11.085
2	1184	675.6	0.650	11.005	10.933–11.083
3	1368	234.9	0.214	10.969	10.904–11.086
4	1343	28.1	0.015	11.061	10.976–11.229

Table 1: Independent benchmark distributions. Costs are billion 2010 USD for the 100-km test coast. Bin ordering matches Figure 1.

bin has 28.1 effective samples and only 1.4 nominal effective samples in a 5% tail. Its median and 95th-percentile bootstrap standard errors are 0.0096 and 0.0199 billion dollars, respectively, without accounting for structural error.

The conditional benchmark medians are nearly flat, unlike the published monotonic increase. Their warmest-to-coolest ratio is about 1.002. Mean protection costs increase from 4.53 to 5.63 billion dollars between these bins, while mean residual flood costs decrease from 5.84 to 4.77 billion dollars. The design rule builds for the maximum water level in each foresight window, so higher early defenses reduce near-term expected flood losses. These terms nearly offset under 4% discounting. This is a consequence of the synthetic geometry, hazard, and decision rule; it is not evidence that global costs are insensitive to warming. Without adaptation, corresponding medians increase from 53.13 to 58.32 billion dollars. Reducing the discount rate to 2% increases the warmest-bin median by approximately 72%.

Warming category	Published 5th	Median	95th	Global error
≤ 1.5	1.0	1.2	1.3	undefined
$(1.5, 2.5]$	1.2	1.5	2.3	undefined
$(2.5, 3.5]$	1.6	2.1	unavailable	undefined
> 3.5	2.2	2.6	unavailable	undefined

Table 2: Published global costs in trillion USD2010, from manuscript prose. Absolute and relative dollar discrepancies are undefined because this project does not estimate the same global quantity. Unavailable values were not digitized.

Published median ratios relative to the coolest bin are 1, 1.25, 1.75, and 2.167. The saved comparison table reports corresponding benchmark ratios and their differences as descriptive shape diagnostics only. Similar ratios would not validate synthetic exposure, and different ratios cannot isolate model error from calibration or accounting differences. The benchmark assigns 1.54% probability to the warmest category, compared with 7.78% of the published sample. That discrepancy also reflects the different emissions measure and historical constraints.

The fixed reference model produces approximately 1.239°C warming in 2010–2019 versus the assessed 1.06°C, and 0.542 YJ ocean heat gain in 1971–2018 versus 0.396 YJ. These positive discrepancies are about 17% and 37%, respectively. Sea-level rise in 1901–2018 is approximately 0.195 m versus 0.20 m (about −2.5%). Agreement of that scalar sum does not validate the component partition or future ice response. The 2025 CO_2 holdout is reported separately in `historical_validation.csv`; it is a short conditional forecast, not a carbon-cycle calibration.

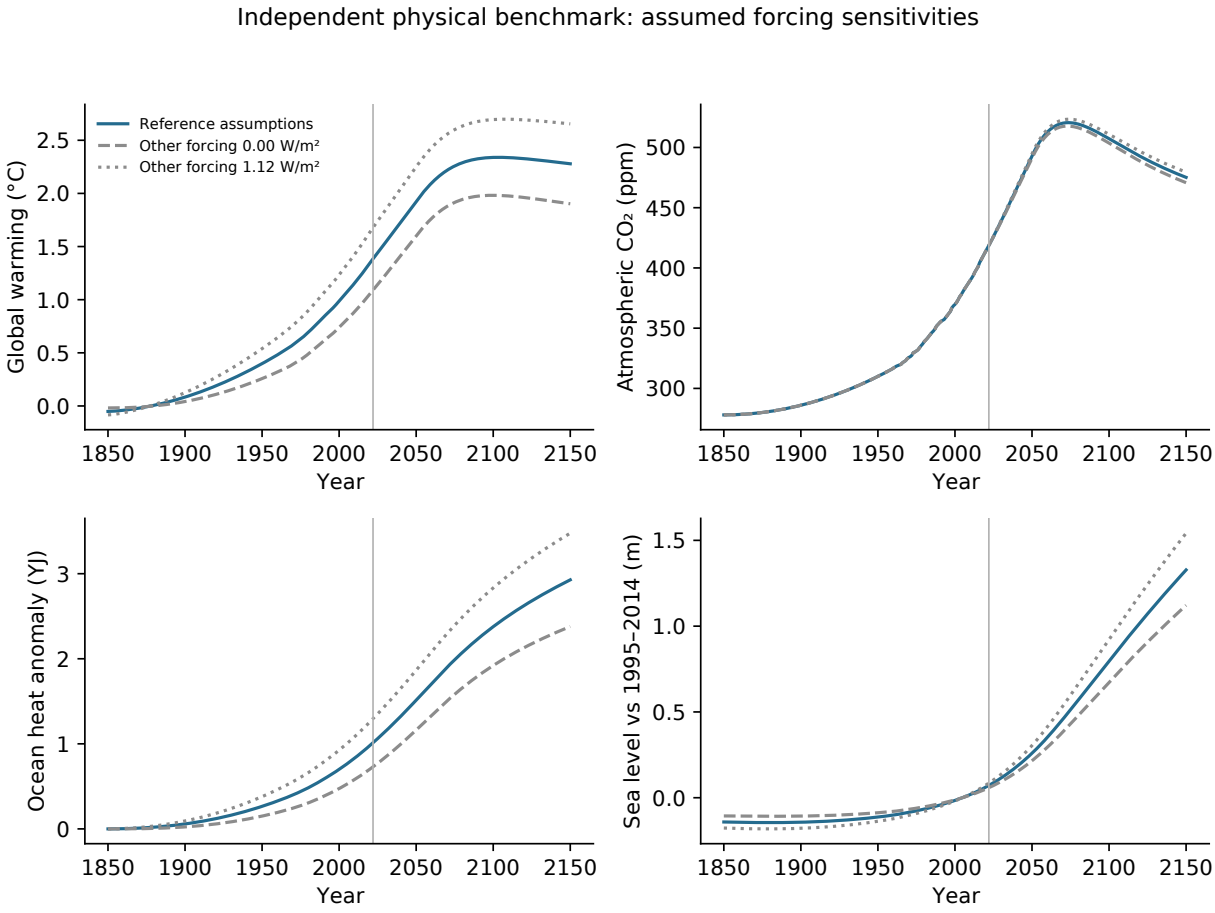


Figure 2: Reference physical trajectories and sensitivity to assumed other forcing of 0, 0.56, and 1.12 W/m^2 after 2019. All curves are independent model output. Temperature uses 1850–1900 and sea level 1995–2014 references; heat is the anomaly from the 1850 initial state. The vertical line marks the 2022 switch from prescribed CO_2 to emissions forcing. Historical prescribed CO_2 agreement is a boundary condition, not a validation result.

5 Verification, limitations, and next steps

The verification executable checks zero forcing, doubled-CO_2 equilibrium, global heat conservation, terrestrial equilibrium, carbon pulse conservation, the reconstructed ocean impulse response, Antarctic reference radius and volume, its volume derivative and mass balance, exact reference-period normalizations, prescribed CO_2, cost accounting, and analytic flood integration against numerical quadrature. The reference Antarctic radius differs from 1.864 million metres by about 0.019%; volume differs from 24.78×10^{15} m^3 by about 0.003%. Carbon budget residuals are of order 10^{-12} Gt C per step, and heat residuals about 10^{-15} YJ per step.

Refinement from a half-year to an eighth-year step at 80 ocean cells changes 2100 warming by about 10^{-5} °C and sea level by 2.3×10^{-5} m. At half-year steps, 40 versus 160 cells changes warming by about 0.0008°C and sea level by 0.0005 m. Cost changes are below 0.01%. Climate and carbon convergence is approximately second order after correcting the historical boundary. The ice model's activation conditions and adaptation choices are nonsmooth, so a single formal convergence order is not claimed for all outputs. The refinement tolerances are 0.02°C, 0.02 m, and 1% cost; they test numerical resolution rather than scientific accuracy.

One initial 4096-proposal run failed the all-members carbonate-domain check. The corrected analysis explicitly records exclusions and their pre-filter weight; it does not conceal the failed check. The primary experiments use one CPU, with no large spatial data products. Source and compact numerical results are retained. The build compiles the C++ model, runs tests and experiments, generates original plots, and compiles this report without network access.

Exact Figure 1 reproduction remains blocked by missing joint calibration and emission probabilities, the unresolved supplement, incomplete forcing and rapid-Antarctic configuration, absent fingerprint and land-motion grids, and unrecovered coastal exposure, hazard, SSP4, and economic parameters. The GTSR archive was identified but could not be retrieved in this environment. These are scientific-input and access limitations, not numerical instabilities. They affect the probability measure and even the definition of the economic quantity, so they cannot be represented by enlarging Monte Carlo error bars.

The existing Git metadata is read-only. Checkpoints were therefore stored in a separate writable Git database inside the project. GitHub API requests fail; the CLI also reports an invalid token for its locally named account. Remote identity, repository existence, and public visibility could not be verified. No public repository or successful push is claimed. Source checkpoints and a portable bundle preserve the work for later publication.

The next scientific step is to obtain the missing written configuration and independent geographic inputs with verified licenses, then calibrate the reconstructed forward system against the specified time series. Only after matching local sea-level mapping, exposure, SSP4, and accounting conventions would a global dollar comparison be interpretable. This benchmark establishes a tested computational starting point and identifies the boundaries of what its present outputs can support.

References

[1] Wong et al. (2025). Coastal adaptation and damage costs at different global warming thresholds. *npj Natural Hazards* 2, 35. doi:10.1038/s44304-025-00089-0.

[2] Urban and Keller (2010). Probabilistic hindcasts and projections of the coupled climate, carbon cycle and Atlantic meridional overturning circulation system. *Tellus A*. doi:10.1111/j.1600-0870.2010.00471.x.

[3] Kriegler (2005). *Imprecise probability analysis for integrated assessment of climate change*. Doctoral dissertation, University of Potsdam. Written source.

[4] Ricciuto, Davis, and Keller (2008). A Bayesian calibration of a simple carbon cycle model. *Global Biogeochemical Cycles* 22, GB2030. doi:10.1029/2006GB002908.

[5] Tanaka (2007). *Inverse estimation for the simple Earth system model ACC2 and its applications*. Doctoral dissertation. Eqs. 2.1.11–12.

[6] Yi et al. (2001). The effects of buffer and temperature feedback on the oceanic uptake of CO_2. *Geophysical Research Letters*. doi:10.1029/2000GL011569.

[7] Fernández Villanueva and Shaffer (2025). DCESS II. *Geoscientific Model Development* 18, 2161–2192. doi:10.5194/gmd-18-2161-2025.

[8] Wong et al. (2017). BRICK v0.2, a simple, accessible, and transparent model framework for climate and regional sea-level projections. *Geoscientific Model Development* 10, 2741–2760. doi:10.5194/gmd-10-2741-2017.

[9] Shaffer (2014). Formulation, calibration and validation of the DAIS model (version 1). *Geoscientific Model Development* 7, 1803–1818. doi:10.5194/gmd-7-1803-2014.

[10] Diaz (2016). Estimating global damages from sea level rise with the Coastal Impact and Adaptation Model (CIAM). *Climatic Change*. doi:10.1007/s10584-016-1675-4.

[11] Wong et al. (2022). Sea level and socioeconomic uncertainty drives high-end coastal adaptation costs. *Earth's Future*. doi:10.1029/2022EF003061.

[12] NOAA Global Monitoring Laboratory (2026). Trends in atmospheric carbon dioxide, annual mean Mauna Loa record. Snapshot dated 9 October 2026. Data source.

[13] IPCC (2021). Climate Change 2021: The Physical Science Basis. Working Group I, Summary for Policymakers and Chapters 7 and 9. Assessment source.